Rui Zhao

Computer Science · Artificial Intelligence

Tsinghua University
Beijing, China
+86 138 1032 7597
zhaorui0329@gmail.com
yituoren.github.io
yituoren

Personal Statement

I am a Computer Science undergraduate at Tsinghua University specializing in AI, with research interests spanning Reinforcement Learning, World Models and Embodied AI, Computer Vision, and Generative Foundation Models. My long-term goal is to integrate high-level symbolic reasoning with low-level embodied perception, working toward generalist agents that can both reason about and physically interact with the open world.

Driven by the conviction that physical-world dynamics cannot be acquired from human-produced text alone — that the embodied side of intelligence requires environments where learning is not bound by physical time — I have moved progressively from symbolic knowledge and reasoning toward generative world models and policy learning.

I see the integration of symbolic, linguistic intelligence with embodied, multi-modal perception as one of the central unsolved problems on the path toward general intelligence, and I hope to contribute to it through both architectural innovation in non-language modalities and the exploration of digitized environments as training grounds.

With research experience and learning ability far beyond my peers, I have a strong foundation in mathematics, artificial neural networks, reinforcement learning, large models and agent. I have a solid mathematical foundation, hands-on experience reproducing and extending complex research codebases, and a track record of proposing ideas and turning them into working systems through extensive experimentation.

I am interested in history, philosophy, literature and music so that I can think about the problems in the world from multiple perspectives and get inspiration from them.

Currently, I am working under the guidance of Prof. Alex Lamb, exploring world models and dynamics understanding. I also have experience working with Postdoc Songjie Niu on knowledge graph construction and with Prof. Hongning Wang on AI teaching assistant and reinforcement learning for cognitive reasoning.

Research Interests

Reinforcement Learning, World Model and Embodied AI, Computer Vision, Generative Models.

Education

2023 – **B.Eng. Department of Computer Science and Technology**, *Tsinghua University*, Beijing, China

Academic Performance

GPA: 3.9/4.0, **Rank**: 9/172

Research Projects

Action-space Model-based Reinforcement Learning for VLA

2026 – now

- Used a bidirectional encoder to replace exploration through the denoising process by traditional gaussian action disturbance, ensuring fine-grained control over the action space
- Designed the algorithm to perturb actions only within recognized key chunks to improve exploration efficiency and reduce redundant rollouts
- Introduced world model to the training loop to provide suffient data and accurate guidance

Compressing Generative World Models via Reinforcement Learning

2025 – now

- Tried to remove the noises and reserve task-relevant or reward-relevant information in world model by making the output more compressed
- Designed a composite reward signal including jpeg compressibility, action prediction consistency (via a custom resnet inverse action model), and value preservation to guide a diffusion world model finetuning process
- Implemented PPO, DDPO, and AWM, adapting and unifying their codebases into a custom pipeline for diffusion world model fine-tuning and conducted extensive experiments to iteratively refine the algorithms based on empirical results
- Focused on efficient and stable RL post-training on world model to improve long-horizon accuracy and downstream policy performance

RL for Reasoning

2025

- Used the concepts in human congnitive psychology to design the reased process
- Constrcuted the cognitive graph / tree to represent the reasoning process and finding the better trajectory
- Used groups of samples with relative scores generated by DeepSeek R1 to train a smaller reward model
- Finetuned Qwen by RL to enhance general cognitive reasoning ability, not only on special tasks or datasets, and achieving better performance

Automated Construction of Knowledge Graph in Knowledge-intensive Domain

2024 – 2025

- Proposed an expandable framework of KG construction Tree-KG, including initial construction and iterative expansion
- Achieved sota performance compared with existing methods such as GraphRAG, especially when using textbooks
- Paper accepted at ACL 2025 main conference, Patent number 202610098472.8

Industrial Experience

2026 – now **Team Member of Embodied AI Business Unit**, *Noetix Robotics*, Beijing, China

Doing research and development on embodied AI, including:

- **World Action Model Pretraining and Post-training**
- **VLA Post-training**: Designing and reproducing memory system and hierarchical inference mechanisms, implementing supervised finetuning and reinforcement learning to train robots to do real-world long-horizon tasks
- **Real-robot Deployment**

2024 – 2025 **AI Teaching Assistant (Developer of AI Cosmos)**, *Tsinghua University*, Beijing, China

Developed the core Meta Agent for the university's AI education system

- **Meta Agent Architecture**: Based on **OpenManus**, designed a high-level agent capable of formulating reusable, generalizable workflows for diverse teaching tasks (e.g., grading, course design) from single instructions
- **Automated Tool & Workflow Construction**: Implemented a "Tool Maker" to synthesize Python scripts as new tools, and a "Workflow Maker" to organize operations into executable graphs, enabling the system to self-expand its capabilities

Also worked on knowledge extraction: Extracting knowledge from textbooks and other materials to build a knowledge graph for the course

Now AI Cosmos has been used in the university's daily academical activities, including course learning and research assistance.

Honors and Scholarships

Program	Tsinghua University Undergraduate Academic Research Promotion Program Level B, 2026
Program	Tsinghua University Undergraduate Academic Research Promotion Program Level A, 2025
Scholarship	Comprehensive Performance Scholarship, 2025
Competition	Second Prize, The 43th Tsinghua University "Challenge Cup" (Information Technology Track), 2025
Competition	10th Place, "Rollman" - The 29th Intelligent Agent Competition (Ghost Track), 2025
Honor	Best Popularity Award, Student Innovation and Research Annual Conference, 2024
Competition	3rd Place, Red Mountain Open-Source Large Model Innovation Competition, 2024
Scholarship	Academic Excellence Scholarship, 2024
Scholarship	ShuPing Scholarship, 91-92, 93-94, 95-96

Skills

Languages	Python, C, C++, Rust, Typescript, Javascript, SystemVerilog
Frameworks	PyTorch, HuggingFace, Weight & Biases, OpenCV
Tools	Git, Docker, Linux, LaTeX, VSCode

Publications

[1] Songjie, N., et al. *Tree-KG: An Expandable KG Construction Framework for Knowledge-intensive Domains.* ACL 2025 main conference.

Languages

Chinese	Native
English	Fluent (CET-6: 611; TOEFL: 112, Reading 30, Listening 30, Speaking 24, Writing 28)